

Unequal Job Opportunities and Well-Being Inequality

A Latent-Jobs Structural Decomposition

Hisham Haydar

University of Luxembourg and LISER

Observed

C_i

Gini of consumption

single adults 0.2633 couples 0.2268

Well-being

M_i

Gini of the attained-bundle money metric

single adults 0.2498 couples 0.1974

low earnings $\Leftarrow$ tastes **or** opportunities **or** resources

equivalised, household-weighted, within each population

Compensation

unequal job opportunities should not become
unequal well-being

Responsibility

preference-based choices should not be
compensated

Compensation vs. Responsibility

$\nexists W : \text{Full Compensation} \wedge \text{Full Responsibility}$

[Fleurbaey & Maniquet]

Research question

How much inequality in money-metric well-being is associated with unequal job opportunities rather than heterogeneous preferences, once labour supply is modelled as choice among latent jobs?

1. Do observed labour-supply choices reflect preferences alone, or also heterogeneous latent job opportunities?
2. How do conclusions differ when welfare evaluates the attained bundle versus the ex-ante opportunity prospect?
3. In a restricted P/A/B decomposition, how much welfare inequality is associated with preferences, local access, and earning opportunities?

Latent jobs

Aaberge, Dagsvik & Strøm

Dagsvik & Jia

Capéau, Decoster & Dekkers

Money metrics

Fleurbaey & Maniquet

Decoster & Haan

Jacquet, Jia & Thoresen

Decomposition

Shorrocks

Creedy & Hérault

Mühlhan

reforms and *changes* → a *level* of well-being inequality

preferences vs. opportunities, inside an estimated latent-jobs model

$j = (e, k, h, w)$

4 occupation groups

$C_i(j)$ priced through EUROMOD

$$v_i(j) = L_i(j) + \beta_c \log\left(\frac{C_i(j)}{\lambda_c}\right)$$

L_i

value of leisure: age; children for
women

β_c

weight on log consumption,
estimated

λ_c

units convention

Opportunities: the choice probability

$$P_i(j) = \frac{\exp\{v_i(j)\} g_i(j)}{\int \exp\{v_i(k)\} g_i(k) d\nu(k)}$$

access

employment mass

hours

bands

occupation

by sex

wage offers

education, experience

valued **or** plentiful

EU-SILC France

collected 2016, incomes 2015

1,540 single-adult households

2,223 couple households

EUROMOD

French 2015 policy system

$h \in [5, 70]$ $w \in [2, 590]$

100 priced alternatives per household

Household-specific opportunities

free coordinates 41

population fit 0.0140

Common opportunities

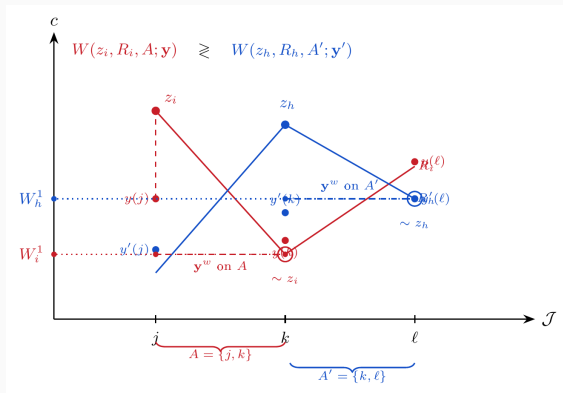
free coordinates 10 / 16

population fit 0.0273 / 0.0264

criterion worse by at least 141.64

choices reflect opportunities, not preferences alone

single adults; population fit is the mean absolute deviation of predicted from observed moments



Own-set equal-consumption equivalents: each individual gets a common consumption on every job in their own ability set; the level at which the preferred job is indifferent to the attained bundle is their money metric, comparable across individuals. Adapted from Haydar and Maniquet (2026), work in progress.

$$J_i = \int e^{L_i(j)} \left(\frac{C_i(j)}{\lambda_c} \right)^{\beta_c} \widehat{g}_i(j) d\nu(j), \quad H_i = \int e^{L_i(j)} \widehat{g}_i(j) d\nu(j),$$

$$M_i^{\text{EA}} = \lambda_c \exp \left\{ \frac{\log J_i - \log H_i}{\beta_c} \right\}$$

J_i

the actual prospect

H_i

same jobs, equal pay

M_i^{EA}

flat-consumption equivalent

$$M_i^{\text{att}} = C_i^{\text{obs}} \exp \left\{ \frac{L_i(j_i^{\text{obs}}) - L_i(o)}{\beta_c} \right\}$$

o

non-employment, reachable by all

j_i^{obs}

the only route for opportunities

Inequality and Shapley decomposition

P

preferences: age, children

A

local unemployment
exposure, region, urban
or rural location, year

B

wage-offer location

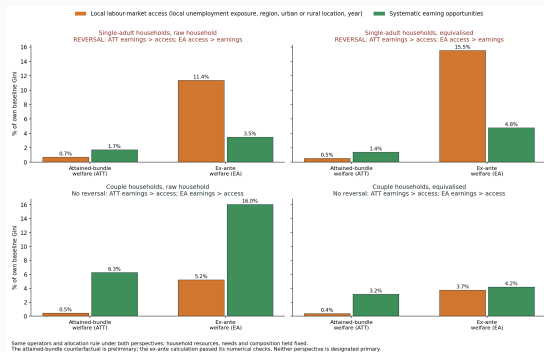
held fixed

household resources,
needs and composition

$$\phi_k^P = \sum_{S \subseteq \{P, A, B\} \setminus \{k\}} \frac{|S|! (3 - |S| - 1)!}{3!} \left[v^P(S \cup \{k\}) - v^P(S) \right]$$

$$v^P(S) = I^P(\emptyset) - I^P(S) \quad I^P: \text{household-weighted Gini of } M^P$$

Results: prospects versus attained outcomes



EA: prospects

access + earnings 7.9–21.3% of baseline Gini
single adults: access > earnings; couples: no
reversal

ATT: attained outcomes

access + earnings 1.9–6.7% of baseline Gini
earnings > access, both household types

What is not claimed

Scope

Household resources, needs and composition
held fixed

Access: local access, not total opportunity

Preliminary decomposition

Not claimed

Not causal

No parameter uncertainty yet

Preferences are not responsibility

Neither perspective is designated primary

prospects versus attained outcomes: two different welfare questions

1. Choices reflect opportunities, not preferences alone.
2. Prospects: access leads for single adults. Attained outcomes: earnings lead. Couples: no reversal.
3. Restricted P/A/B: a bounded part of well-being inequality; preference sign not robust.

next: resources and needs as a fourth channel · parameter uncertainty · validation

Backup: ex-ante decomposition

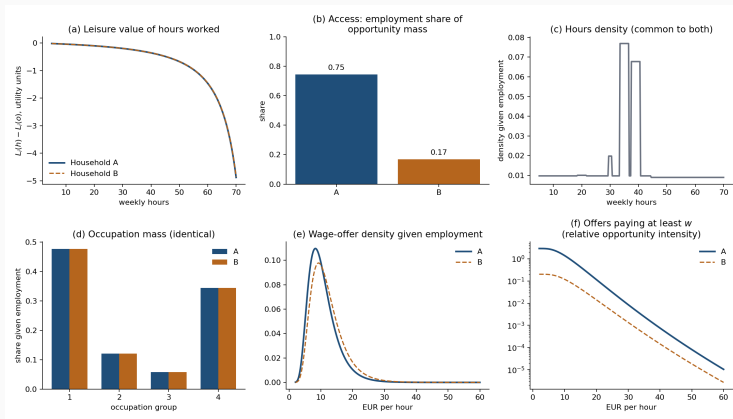
	scale	access Gini points	earnings Gini points	access + earnings % of baseline	ordering
Single adults	raw	0.01812	0.00555	14.8	access > earnings
	equivalised	0.02183	0.00673	20.3	access > earnings
Couples	raw	0.00667	0.02049	21.3	earnings > access
	equivalised	0.00449	0.00504	7.9	earnings > access

single adults: access = $3.3 \times$ earnings (raw), $3.2 \times$ (equivalised)

Backup: attained-bundle decomposition

	scale	access Gini points	earnings Gini points	access + earnings % of baseline	ordering
Single adults	raw	0.0016	0.0040	2.4	earnings > access
	equivalised	0.0013	0.0034	1.9	earnings > access
Couples	raw	0.0009	0.0128	6.7	earnings > access
	equivalised	0.0007	0.0062	3.5	earnings > access

Backup: matched households



access: $A = 14.5 \times B$

wage-offer location: $B + 11.5$ log points

Backup: equivalisation

Couples, ex ante: access + earnings

21.3% $\longrightarrow$ 7.9%

raw $\rightarrow$ equivalised

Couples, ex ante: preferences

0.00291 $\longrightarrow$ -0.00212

Gini points, raw $\rightarrow$ equivalised

modified-OECD scale: a convention, reported both ways